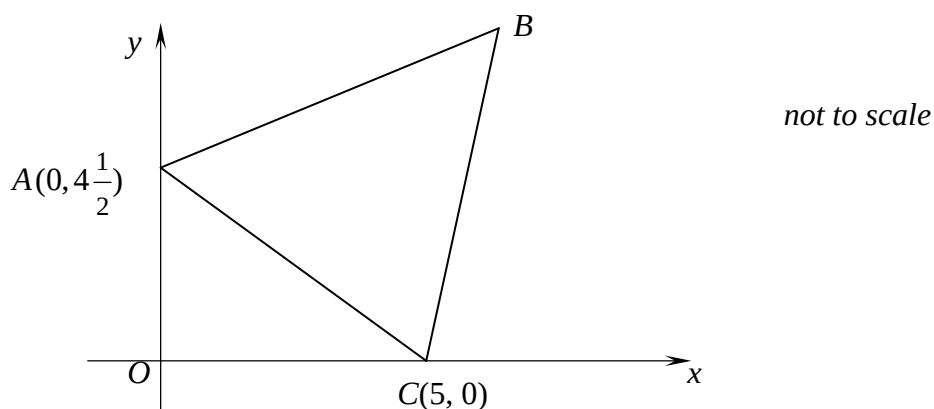


Question 1.**Marks**

- (a) Evaluate $\frac{e^3 - 2 \cdot 1^2}{\sqrt{3 \cdot 14 + 2 \cdot 1}}$ correct to 2 significant figures. **1**
- (b) Given that $\tan \theta = \frac{7}{8}$ and $\cos \theta < 0$, find the exact value of $\operatorname{cosec} \theta$. **2**
- (c) Solve for x : $|4x - 15| \leq 3$. **2**
- (d) State the period and the amplitude for the graph of $3y = \sin\left(2x - \frac{\pi}{4}\right)$. **2**
- (e) Paint at the local hardware store is sold at a profit of 30% on the cost price. If a can of paint is sold for \$67.60, find the cost price. **2**
- (f) Solve for α : $\tan \alpha = -0.5$, where $0 < \alpha < \pi$, correct to 2 decimal places. **1**
- (g) Two fair dice are rolled at random. Find the probability that the two numbers are the two digits of a perfect square? **2**

Question 2.**[START A NEW PAGE]**

The lines AB and CB have equations: $x - 2y + 9 = 0$ and $4x - y - 20 = 0$ respectively.

- (a) Show that the equation of the line AC is $9x + 10y - 45 = 0$. **2**
- (b) Calculate the exact distance AC . **1**
- (c) Find the coordinates for point B . **2**
- (d) Find the angle of inclination of the line through A and B (to nearest degree). **2**
- (e) Calculate the shortest distance from point B to the line AC . Hence find the area of triangle ABC . **2**
- (f) Determine the inequalities that define the area bounded by $\triangle ABC$. **3**

Question 3.	[START A NEW PAGE]	Marks
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- | | | |
|-----|---|---|
| (a) | Find $\frac{dy}{dx}$ for | |
| | (i) $y = (1 + \ln x)^2$. | 2 |
| | (ii) $y = \frac{\sin x}{e^{3x} + 1}$. | 2 |
| (b) | (i) Find $\int (5x - 1)^3 dx$. | 2 |
| | (ii) Find the value: $\int_0^{\pi} \sec^2 \frac{x}{4} dx$. | 2 |
| (c) | Given that α and β are the roots of the equation $2x^2 - 6x - 7 = 0$, | |
| | (i) Find $\alpha + \beta$. | 1 |
| | (ii) Find $\alpha^2 + \beta^2$. | 2 |
| | (iii) Find $\alpha^2 - 3\alpha$. | 1 |

Question 4.	[START A NEW PAGE]
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- | | | |
|-----|--|---|
| (a) | The good ship Lollypop sails from port A 60 nautical miles due west to port B. It then sails a distance of 50 nautical miles on a bearing of 210° T to port C | |
| | (i) Draw a diagram to illustrate this information. | 1 |
| | (ii) Calculate the distance of C from A (to nearest nautical mile). | 2 |
| | (iii) Calculate the bearing of port C from port A. | 2 |
| (b) | Evaluate: $\lim_{x \rightarrow 16} \frac{x - 16}{\sqrt{x} - 4}$. | 2 |
| (c) | Find the equation of the normal to the curve $y = 1 + \ln 2x$ at the point $\left(\frac{e}{2}, 2\right)$. | 3 |
| (d) | Find the radius and the centre of the circle whose equation is: | 2 |

$$4x^2 - 4x + 4y^2 + 24y + 21 = 0.$$

Question 5.

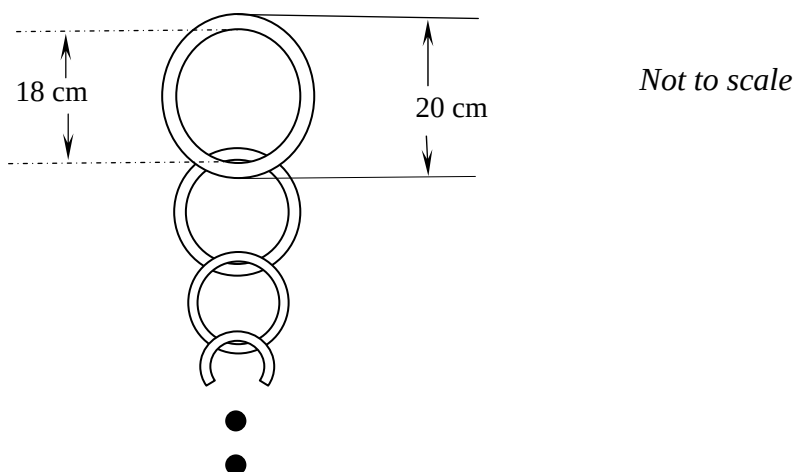
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Marks

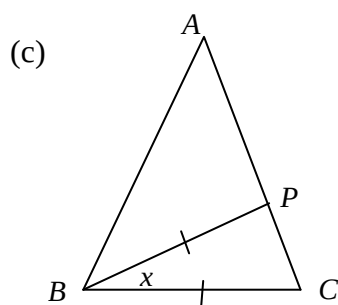
- (a) 1 000 tickets are sold in a raffle. First prize is \$1 000, second prize is \$500 and third prize of \$200.
The prize winning tickets are drawn consecutively without replacement where the first ticket wins first prize.

- (i) Find the probability that a person buying one ticket in the raffle wins:
- | | |
|---------------------|----------|
| (α) First prize. | 1 |
| (β) at least \$500. | 1 |
| (γ) No prizes. | 1 |
- (ii) A person buying two tickets in the raffle wins at least \$500. **2**

- (b) A number of linked rings, each 1 cm thick, are hung from a nail on a wall. The top ring has an outside diameter of 20 cm as shown in the diagram. The outside diameter of each of the other rings is 1 cm less than that of the ring above it. The bottom (last) ring has an outside diameter of 3 cm.



- (i) **Copy** the diagram and explain why the top of the second ring is 17 cm above the top of the third ring. **1**
- (ii) Hence calculate the distance from the top of the top ring to the bottom of the bottom ring. **3**



Not to scale

- Given $\triangle ABC$ is an isosceles triangle with $AB = AC$. **3**
 P lies on AC such that $\angle ABP = 3\angle PBC$ and $BP = BC$.

Copy the diagram into your writing booklet and by letting $\angle CBP = x$, or otherwise, find angle $\angle CBP$ expressing it in radians.

Question 6.**[START A NEW PAGE]****Marks**

(a) Show that $\frac{1 + \tan^2 A}{\operatorname{cosec}^2 A} = \tan^2 A.$ **2**

(b) Given the function $y = f(x)$ for $1 \leq x \leq 2.5$, where **3**

x	1	1.25	1.5	1.75	2	2.25	2.5
$f(x)$	3.4	2.2	0.4	1.9	-2.7	1.3	2

Use Simpsons' rule to evaluate $\int_1^{2.5} f(x) dx$, correct to 1 decimal place, using the 7 function values in the table.

(c) A function is defined by $y = g(x)$, where $g(x) = x^3 - 6x^2 + 5 = (x-1)(x^2 - 5x - 5).$

(i) Determine the coordinates of the stationary points and determine their nature. **3**

(ii) State at what point does the curve change its concavity? **1**

(iii) Hence sketch the graph of $y = g(x)$, showing all essential details. **2**

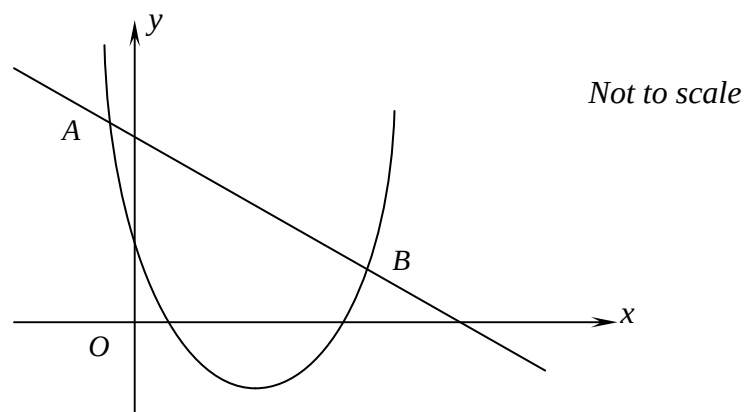
(iv) Find the minimum value of $x^3 - 6x^2 + 5$ when $-3 \leq x \leq 5$. **1**

Question 7.

[START A NEW PAGE]

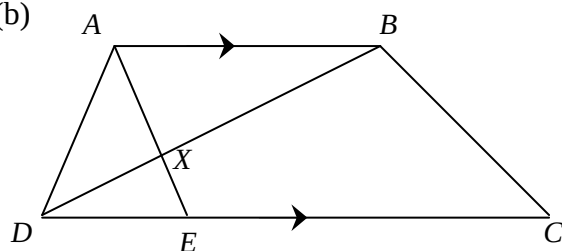
Marks

- (a) The parabola $y = (x - 3)^2 - 1$ and the line $y = 13 - 2x$ intersect at points A and B.



- (i) Find the x -coordinate for points A and B. 2
- (ii) Hence, or otherwise find the area bounded by the parabola and the line. 2

(b)



$ABCD$ is a trapezium where $AB \parallel DC$.
Also $BD \perp BC$ and $AE \perp BD$ at X .

- (i) **Copy** the diagram into your writing booklet and find AX ,
given $AD = 41$ cm, $DX = 9$ cm. 1
- (ii) What type of quadrilateral is $ABCE$? Give reasons. 2
- (iii) Show that $\triangle DXE \parallel \triangle DBC$. 2
- (iv) Hence show that $BX \cdot XE = 360$. 3

Question 8.

[START A NEW PAGE]

Marks

- (a) A function is defined by the following properties:

3

$$y = 0 \text{ when } x = 1; \quad \frac{dy}{dx} = 0 \text{ when } x = -3, 1 \text{ and } 5;$$

$$\text{and } \frac{d^2y}{dx^2} > 0 \text{ for } x < -1 \text{ and } 1 < x < 3.$$

Sketch a possible graph of the function.

- (b) Larsen begins his retirement with \$500 000 at the beginning of 2009. The annual interest rate is 8% *p.a.* Interest is calculated annually on the balance at the beginning of the year and is added to the remaining balance. Larsen plans to withdraw \$56 000 annually, with the first withdrawal at the end of 2009.

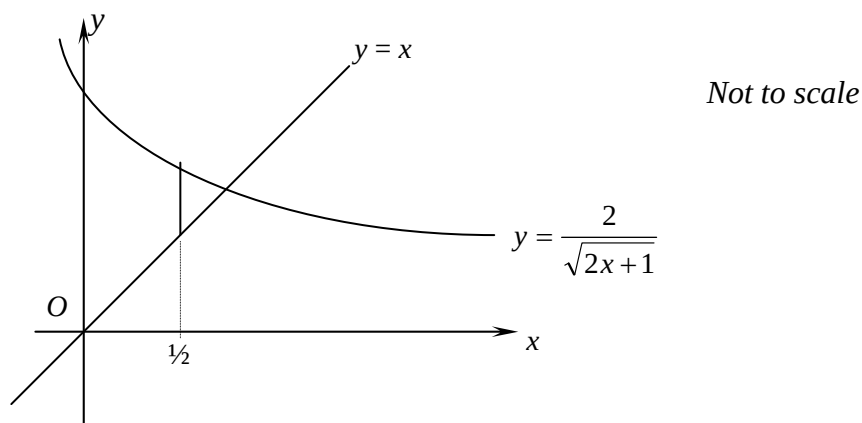
By letting A_n be the remaining balance after the n th withdrawal,

(i) Show that: $A_2 = 5 \times 10^5 R^2 - 5 \cdot 6 \times 10^4 [1 + R]$, where $R = 1 \cdot 08$. **2**

(ii) Hence deduce that: $A_n = 10^5 [7 - 2R^n]$ **2**

(iii) Calculate during which year will Larsen's fund reach zero? **2**

- (c)



The area bounded by the curves $y = \frac{2}{\sqrt{2x+1}}$, $y = x$, the lines $x = 0$ and $x = \frac{1}{2}$ **3**

is rotated about the x -axis.

Find the volume of the solid of revolution formed.

Question 9.	[START A NEW PAGE]	Marks
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(a)	Solve for x : $x^2 - 2 x - 15 = 0$.	2
-----	---	---

(b)	If $\log_a(xy^3) = 1$ and $\log_a(x^2y) = 1$. What is the value of $\log_a(xy)$?	2
-----	--	---

(c)	The rate at which carbon dioxide (CO_2) will be produced when conducting an experiment is given by $\frac{dV}{dt} = 0.01(30t - t^2)$, where $V \text{ cm}^3$ is the volume of the gas at t minutes.	
-----	---	--

(i)	At what rate is the gas being produced 15 minutes after the experiment begins?	1
-----	--	---

(ii)	How much carbon dioxide has been produced during this time?	2
------	---	---

(d)	The mass S grams of a substance decays at an instantaneous rate proportional to its mass present at time t years.	
-----	---	--

(i)	Explain why the rate equation is given by: $\frac{dS}{dt} = -kS$, where $k (> 0)$ is the decay rate constant of proportionality.	1
-----	--	---

(ii)	Verify that $S = S_0 e^{-kt}$ is the general solution to the rate equation, where S_0 is the initial mass of the substance.	1
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(iii)	Show that the half-life for the substance is given by $\frac{\ln 2}{k}$ years	1
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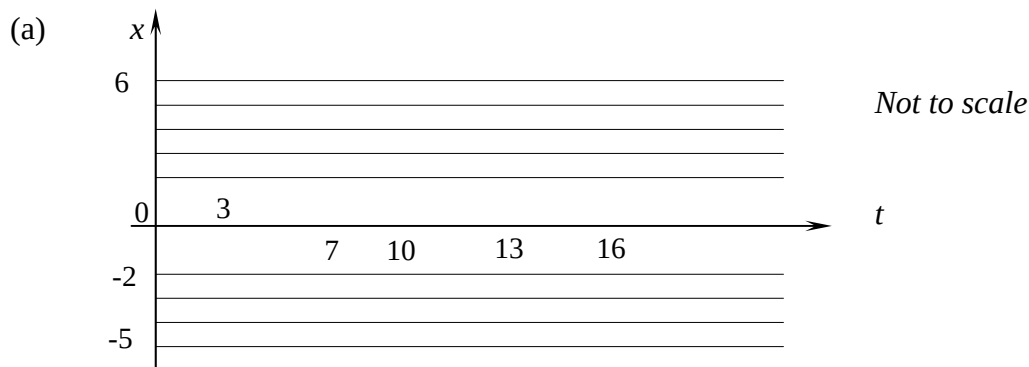
(iv)	If the mass present at time t_1 is S_1 and the mass at time t_2 is S_2 , show that	2
------	--	---

$$k = \frac{1}{t_2 - t_1} \ln \left(\frac{S_1}{S_2} \right), \text{ assuming } t_2 > t_1.$$

Question 10.

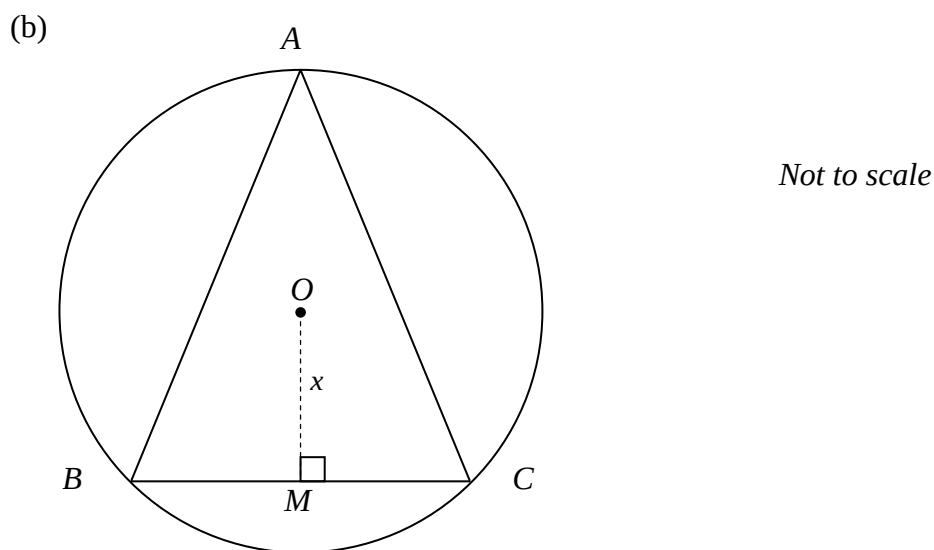
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Marks



A particle is moving along a straight line according to the sketch of displacement x metres against time t seconds above.

- | | | |
|-------|---|---|
| (i) | When is the particle at rest? | 1 |
| (ii) | At what time does the particle have greatest speed (approximately)? | 1 |
| (iii) | Describe what happens to the particle as $t \rightarrow \infty$. | 1 |
| (iv) | Calculate the distance that the particle has eventually travelled. | 2 |



An isosceles triangle ABC with $AB = AC$ is inscribed in a circle centre O and of radius R units.

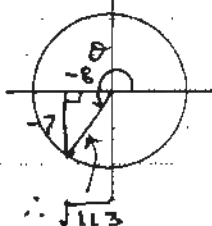
Given that $OM = x$ units, $OM \perp BC$ and M is the midpoint of BC ,

- | | | |
|------|--|---|
| (i) | Show that the area of $\triangle ABC$, S square units, is given by: | 2 |
| | $S = (R + x)\sqrt{R^2 - x^2}.$ | |
| (ii) | Hence show that the triangle with maximum area is an equilateral triangle. | 5 |

20 MATHEMATICS TRIAL, 2009.

MARKING SCHEME

MATHEMATICS: Question 1

Suggested Solutions	Marks	Marker's Comments																																				
1(a) $\frac{2^3 - 2 \cdot 1^2}{\sqrt{3 \cdot 14 + 2 \cdot 1}} = \frac{2^3 - 4 \cdot 1}{\sqrt{5 \cdot 24}} = \frac{6 \cdot 8478901...}{\sqrt{120}} = 6.8 \quad (2 \text{ SF}) \checkmark$		[1]																																				
(b) $\tan \theta = \frac{1}{8} \quad \cos \theta < 0 \Rightarrow 3^{\text{rd}} \text{ Quadrant}$ $\cos \theta = \frac{1}{\sin \theta} = -\frac{\sqrt{113}}{7}$  $\therefore \sqrt{113}$		[2]																																				
(c) $4x - 15 \leq 3$ $\therefore -3 \leq 4x - 15 \leq 3$ $12 \leq 4x \leq 18$ $3 \leq x \leq 4.5 \quad \checkmark$		[2]																																				
(d) $3y = \sin(2x - \frac{\pi}{4})$ $\therefore y = \frac{1}{3} \sin(2x - \frac{\pi}{4})$ Period = $\frac{2\pi}{2} = \pi$ Amplitude = $\frac{1}{3}$		[2]																																				
(e) SELLING = $130 \cdot 6 \text{ cost} = \\67.60 $\therefore \text{cost} = \frac{\\$67.60 \times 100}{130}$ $= \\$52 \quad \checkmark$		[2]																																				
(f) $\tan \alpha = -0.5 \quad 0 < \alpha < \pi \quad 2^{\text{nd}} \text{ Q.}$ $\alpha = \pi - 0.4676...$ $= 2.677945...$ $\alpha = 2.68 \quad (2 \text{ dp}) \quad \checkmark$		[1]																																				
(g) <table border="1" style="width: 100%; text-align: center;"> <tr><td>11</td><td>21</td><td>31</td><td>41</td><td>51</td><td>61</td></tr> <tr><td>12</td><td>22</td><td>32</td><td>42</td><td>52</td><td>62</td></tr> <tr><td>13</td><td>23</td><td>33</td><td>43</td><td>53</td><td>63</td></tr> <tr><td>14</td><td>24</td><td>34</td><td>44</td><td>54</td><td>64</td></tr> <tr><td>15</td><td>25</td><td>35</td><td>45</td><td>55</td><td>65</td></tr> <tr><td>16</td><td>26</td><td>36</td><td>46</td><td>56</td><td>66</td></tr> </table> $S =$ $P(E = \text{perfect eq. digits}) = \frac{8}{36} = \frac{2}{9} \quad \checkmark$	11	21	31	41	51	61	12	22	32	42	52	62	13	23	33	43	53	63	14	24	34	44	54	64	15	25	35	45	55	65	16	26	36	46	56	66		[2]
11	21	31	41	51	61																																	
12	22	32	42	52	62																																	
13	23	33	43	53	63																																	
14	24	34	44	54	64																																	
15	25	35	45	55	65																																	
16	26	36	46	56	66																																	

MATHEMATICS: Question 2

Suggested Solutions	Marks	Marker's Comments
Q 2(a) $m_{AC} = \frac{4\frac{1}{2} - 0}{0 - 5} = -\frac{4\frac{1}{2}}{5} = -\frac{9}{10}$ ✓ Eqn. of AC : $y = mx + b$ $y = -\frac{9}{10}x + 4\frac{1}{2}$ ✓ $10y = -9x + 45$ $\therefore 9x + 10y - 45 = 0$ qed		2
b) $AC^2 = 5^2 + 4\frac{1}{2}^2$ (Pyth Thm) $AC^2 = \frac{181}{4}$ $\therefore AC = \frac{\sqrt{181}}{2}$ units ✓		1
(c) $x - 2y + 9 = 0$ — (1) : AB $4x - y - 20 = 0$ — (2) : CB $\therefore \frac{8x - 2y - 40 = 0}{7x - 4y = 0}$ — (2a) $\therefore x = 7$ ✓ Subst in (2) $28 - y - 20 = 0$ $\therefore y = 8$ ✓ $\therefore B = (7, 8)$		2
(d) $m_{AB} = \frac{1}{2}$ ✓ from $x - 2y + 9 = 0$ $\tan \theta = \frac{1}{2}$ $\angle \theta = 26^\circ 34' \div 27^\circ$ ✓		2
(e) AC : $9x + 10y - 45 = 0$ B : (7, 8) $\perp$ dist = $\frac{ 9 \times 7 + 10 \times 8 - 45 }{\sqrt{9^2 + 10^2}} = \frac{98}{\sqrt{181}}$ ✓ $\therefore \text{AREA } \triangle ABC = \frac{1}{2} \times AC \times \perp d.$ $= \frac{1}{2} \times \frac{\sqrt{181}}{2} \times \frac{98}{\sqrt{181}} = 24\frac{1}{2} \text{ u}^2$ ✓		2
(f) AC : $9x + 10y - 45 \geq 0$ BC : $4x - y - 20 \leq 0$ AB : $x - 2y + 9 \geq 0$		3

MATHEMATICS: Question 3

Suggested Solutions	Marks	Marker's Comments
Q3(a) (i) $y = (1 + \ln x)^2$ $\frac{dy}{dx} = 2(1 + \ln x) \times \frac{1}{x} = \frac{2(1 + \ln x)}{x}$ ✓ (ii) $y = \frac{\sin x}{e^{3x} + 1}$: $\frac{dy}{dx} = \frac{\cos x(e^{3x} + 1) - \sin x \times 3e^{3x}}{(e^{3x} + 1)^2}$ ✓✓ $\frac{dy}{dx} = \frac{(e^{3x} + 1) \cos x - 3 \sin x \cdot e^{3x}}{(e^{3x} + 1)^2}$		[2] [2]
(b) (i) $\int (5x - 1)^3 dx = \frac{1}{4 \times 5} (5x - 1)^4 + C$ ✓✓ $= \frac{1}{20} (5x - 1)^4 + C$ (ii) $\int_0^{\pi} \sec^2 \frac{x}{4} dx = 4 \left[\tan \frac{x}{4} \right]_0^{\pi}$ ✓ $= 4 \left[\tan \frac{\pi}{4} - \tan 0 \right]$ $= 4 [1 - 0] = 4$ ✓		[2] [2]
(c) (i) $2x^2 - 6x - 7 = 0$ $\therefore \alpha + \beta = -\frac{b}{a} = -\frac{(-6)}{2} = 3$ ✓ (ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ ✓ $= 3^2 - 2 \times \left(-\frac{7}{2} \right) = 9 + 7$ $= 16$ ✓ or $2\alpha^2 - 6\alpha - 7 = 0$ as α is a root $2\beta^2 - 6\beta - 7 = 0$ $2(\alpha^2 + \beta^2) - 6(\alpha + \beta) - 14 = 0$ $2(\alpha^2 + \beta^2) - 6 \times 3 - 14 = 0$ $\therefore \alpha^2 + \beta^2 = \frac{32}{2} = 16$ (iii) As α is a root $2\alpha^2 - 6\alpha - 7 = 0$ $2\alpha^2 - 6\alpha = 7$ $\therefore \alpha^2 - 3\alpha = \frac{7}{2} = 3\frac{1}{2}$ ✓		[1] [2] [1]

MATHEMATICS: Question 4

Suggested Solutions	Marks	Marker's Comments
Q 4(a) (i) $\angle ABC = 120^\circ$ $\angle CAB = 120^\circ$		1
(ii) $AC^2 = 50^2 + 60^2 - 2 \times 50 \times 60 \times \cos 120^\circ$ $= 9100$ $AC = 10\sqrt{91} = 95.39392014...$ $\therefore AC = 95 \text{ n.miles (nearest nm)}$		2
(iii) Let $\angle CAB = \theta$ $\frac{\sin \theta}{50} = \frac{\sin 120^\circ}{AC}$ $\sin \theta = \frac{50 \times \sin 120^\circ}{AC} = \begin{cases} 0.4539206... \\ 0.455802... \end{cases}$ $\angle \theta = \begin{cases} 27^\circ \text{ or } 153^\circ \\ 27^\circ 7' \text{ or } 152^\circ 53' \end{cases}$ but $120^\circ + 152^\circ > 180^\circ$ $\therefore \angle \theta = 27^\circ \text{ or } 27^\circ 7'$ $\therefore$ Bearing of C from A is $243^\circ \text{ T} / 242^\circ 53' \text{ T}$		2
(b) $\lim_{x \rightarrow 16} \frac{x-16}{\sqrt{x}-4} = \lim_{x \rightarrow 16} \frac{(\sqrt{x}-4)(\sqrt{x}+4)}{\sqrt{x}-4}$ $= \lim_{x \rightarrow 16} (\sqrt{x}+4) = \sqrt{16}+4 = 8$	(or) $\frac{x-16}{\sqrt{x}-4} \times \frac{\sqrt{x}+4}{\sqrt{x}+4} = \frac{(x-16)(\sqrt{x}+4)}{x-16} = \sqrt{x}+4$	2
(c) $y = 1 + \ln 2x$ $\frac{dy}{dx} = \frac{1}{2x} \times 2 = \frac{1}{x}$ Gradient of Tangent at $x = \frac{1}{2}e$: $m_T = \frac{2}{e}$ " of Normal $m_N = -\frac{e}{2}$ Eqn. of Normal at $(\frac{1}{2}e, 2)$: $y-2 = -\frac{e}{2}(x-\frac{1}{2}e)$ $\therefore y = -\frac{1}{2}ex + \frac{1}{4}e^2 + 2$	$2ex + 4y - e^2 - 8 = 0$	3
(d) $4x^2 - 4x + 4y^2 + 24y + 21 = 0$ $x^2 - x + \frac{1}{4} + y^2 + 6y + 9 = -\frac{21}{4} + \frac{1}{4} + 9$ $(x - \frac{1}{2})^2 + (y + 3)^2 = 4$ $\therefore$ centre is $(\frac{1}{2}, -3)$ ✓ radius = 2		2

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MATHEMATICS: Question 6

6

Suggested Solutions

Marks

Marker's Comments

26(a) LHS = $\frac{1 + \tan^2 A}{\cos^2 A} = \frac{\sec^2 A}{\cos^2 A}$ ✓

Approach I:

$$= \frac{1}{\cos^2 A} \div \frac{1}{\sin^2 A}$$

$$= \frac{\sin^2 A}{\cos^2 A}$$

$$= \tan^2 A \quad \text{✓}$$

II: $\frac{1 + \tan^2 A}{\cos^2 A} = \frac{\sin^2 A (1 + \frac{1}{\sin^2 A})}{\cos^2 A}$

$$= \frac{\sin^2 A}{\cos^2 A} (1 + \frac{1}{\sin^2 A})$$

$$= \frac{\sin^2 A}{\cos^2 A} (\frac{\sin^2 A + 1}{\sin^2 A})$$

$$= \frac{\sin^2 A + 1}{\cos^2 A} = \frac{\sec^2 A}{\cos^2 A} = \tan^2 A$$

2

(b)

x	1	1.25	1.5	1.75	2	2.25	2.5
f(x)	3.4	2.2	0.4	1.9	-2.7	1.3	2

$y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6 \quad y_7$

$$h = \frac{2.5 - 1}{6} = \frac{1.5}{6} = \frac{1}{4} = 0.25 \quad \checkmark$$

$$\int f(x) dx \approx A_3 \approx \frac{h}{3} [y_1 + y_7 + 4(y_2 + y_4 + y_6) + 2(y_3 + y_5)]$$

$$= \frac{0.25}{3} [5.4 + 4(2.2 + 1.9 + 1.3) + 2(0.4 + -2.7)]$$

$$= \frac{0.25}{3} [5.4 + 4 \times 5.4 + 2 \times (-2.3)]$$

$$= \frac{0.25}{3} [5.4 + 21.6 - 4.6] = \frac{0.25}{3} \times 22.4$$

$$= 1.86$$

$$= 1.9 \quad (1 \text{ dp})$$

3

(c) $g(x) = x^3 - 6x^2 + 5 = (x-1)(x^2 - 5x - 5)$

(i) $g'(x) = 3x^2 - 12x = 3x(x-4)$

$g''(x) = 6x - 12 = 6(x-2)$

For SPs to occur $g'(x) = 0$

i.e. $3x(x-4) = 0$

$x = 0$ or 4

$\therefore g(0) = 5$ or -27

$\therefore$ SPs are $(0, 5)$ and $(4, -27)$

TEST: at $x=0$

$g''(0) = -12 < 0$

$\therefore$ concave downwards

$\therefore$ Rel. max TP at $(0, 5)$

at $x=4$

$g''(4) = 12 > 0$

$\therefore$ concave upwards

$\therefore$ Rel. min TP at $(4, -27)$

3

(ii) change in concavity can be at $g''(x) = 0$ P.O.I.

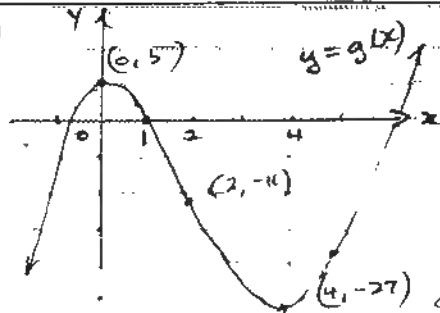
so at $x=2$

$\therefore (2, -11) \quad \checkmark$

1

x	1	2	3
g''(x)	-6	0	6

(iii)



2

(iv) $-3 \leq x \leq 5$

$g(-3) = -76$

$g(5) = -20$

$\therefore$ From sketch $y = g(x)$

Absolute min is -76 at $x = -3$

1

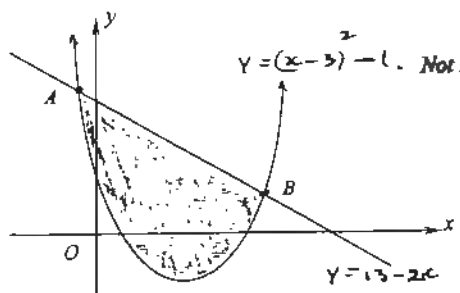
MATHEMATICS: Question 7.

Suggested Solutions

Marks

Marker's Comments

Q7(a) $y = (x-3)^2 - 1$
 $y = 13 - 2x$



(i) POINTS OF INTERSECTION

$$(x-3)^2 - 1 = 13 - 2x$$

$$x^2 - 6x + 9 - 1 = 13 - 2x$$

$$x^2 - 4x - 5 = 0$$

$$(x+1)(x-5) = 0$$

$$\therefore x = -1 \text{ or } 5$$

$$x_A = -1 \quad x_B = 5$$

[2]

(ii) Area = $\int_{-1}^5 (y_U - y_L) dx$

$$= \int_{-1}^5 (13 - 2x - \{(x-3)^2 - 1\}) dx$$

$$= \int_{-1}^5 (14 - 2x - (x-3)^2) dx = \int_{-1}^5 (5 + 4x - x^2) dx$$

$$= \left[5x - x^2 - \frac{1}{3}(x-3)^3 \right]_{-1}^5 = \left[5x + 2x^2 - \frac{1}{3}x^3 \right]_{-1}^5$$

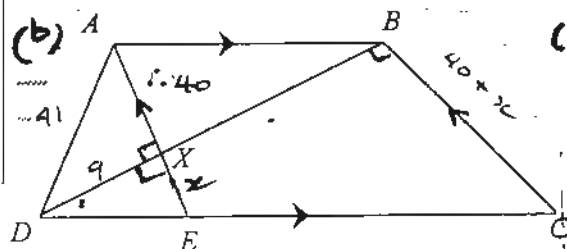
$$= \left[\left(70 - 25 - \frac{8}{3} \right) - \left(-14 - 1 + \frac{64}{3} \right) \right] = 78 - 42$$

$$= 42\frac{1}{3} - 6\frac{1}{3}$$

$$\text{Area} = 36 \text{ sq. units}$$

$$= 36$$

[2]



(i) $41^2 = 9^2 + Ax^2$ (Pyth. Thm.)

$$Ax = 40 \text{ cm}$$

[1]

(ii) As $\angle DXE = \angle DBC = 90^\circ$ and are corresponding angles ✓
 $\therefore AE \parallel CB$ and as $AB \parallel DC$
 $\therefore ABCE$ is a parallelogram ✓ (two pair of parallel sides)

[2]

(iii) In $\Delta s DXE$ and DBC

1. $\angle XDE = \angle BDC$ (common) ✓

2. $\angle DXE = \angle DBC = 90^\circ$ (data)

$\therefore \Delta DXE \parallel \Delta DBC$ (equiangular / matching angles equal) ✓

[2]

(iv) $AE = 40 + xE$

but $BC = AE$ (opp. sides of ||ogram are equal) ✓

$\therefore \frac{BC}{XE} = \frac{DB}{DX}$ (corresponding sides in similar triangles are in the same ratio)

or $\frac{XE}{BC} = \frac{DX}{DB}$

$$\frac{XE}{40+XE} = \frac{9}{9+BX}$$

i.e. $\frac{40+XE}{XE} = \frac{9+XB}{9}$ ✓

$$360 + 9XE = 9XE + XB \cdot XE$$

$$\therefore BX \cdot XE = 360 \text{ qed.}$$

[3]

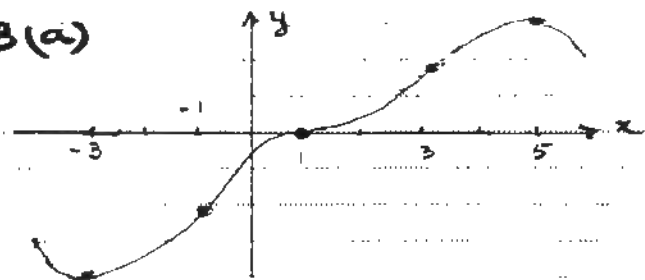
MATHEMATICS: Question 8

Suggested Solutions

Marks

Marker's Comments

Q8(a)



Possible sketch.

Rel min TP $x = -3$ ✓
 Rel max TP $x = 5$ ✓
 P.O.I. at $x = -1, 3$ ✓
 HPOI at $(1, 0)$ ✓

3

b) $r = 8\% = 0.08$
 $A = 1 + \frac{r}{100} = 1.08$ given

(i) Begin 2010 $A_1 = 500000 \times A - 56000$ ✓
 $=$
 Begin 2011 $A_2 = A_1 A - 56000$
 $= (5 \times 10^5 A - 5.6 \times 10^4) A - 5.6 \times 10^4$ ✓
 $= 5 \times 10^5 A^2 - 5.6 \times 10^4 (1 + A)$ qed

2

(ii) so Begin 2012 $A_3 = 5 \times 10^5 A^3 - 5.6 \times 10^4 [1 + A + A^2]$
 $\therefore$ After n years
 $A_n = 5 \times 10^5 A^n - 5.6 \times 10^4 [1 + A + A^2 + \dots + A^{n-1}]$ ✓
 $= 5 \times 10^5 A^n - 5.6 \times 10^4 \left[\frac{A^n - 1}{A - 1} \right]$ ✓
 $= 5 \times 10^5 A^n - 5.6 \times 10^4 \left[\frac{A^n - 1}{0.08} \right]$; $A - 1 = 1.08 - 1$
 $= 5 \times 10^5 A^n - 7 \times 10^5 (A^n - 1)$ ✓
 $= 5 \times 10^5 A^n - 7 \times 10^5 A^n + 7 \times 10^5$ ✓
 $\therefore A_n = 10^5 [7 - 2A^n]$ qed

2

(iii) when $A_n = 0$
 $\therefore 7 - 2A^n = 0$
 $1.08^n = \frac{7}{2} = 3.5$ ✓
 $n = \frac{\ln 3.5}{\ln 1.08} = 16.27788 \dots$

$\therefore$ Funds runs out during 17th year / (but actually at end of) 2025 ✓

2

(c) Vol $= \pi \int_{-1}^2 y_U^2 dx - \pi \int_{-1}^2 y_L^2 dx$
 $= \pi \int_{-1}^2 \left(\frac{4}{2x+1} - x^2 \right) dx$ ✓
 $= \pi \left[2 \ln(2x+1) - \frac{1}{3} x^3 \right]_0^2$ ✓
 $= \pi \left[2 \ln 5 - \frac{8}{3} - 0 \right]$

Vol $= \pi \left(\ln 4 - \frac{1}{3} \right)$ units³ ✓

3

MATHEMATICS: Question 10

Suggested Solutions

Marks

Marker's Comments

Q10(a).

(i) At rest when $v = \frac{dx}{dt} = 0$ $\therefore$ at $t = 3$ and 10 (3 and 10 seconds)

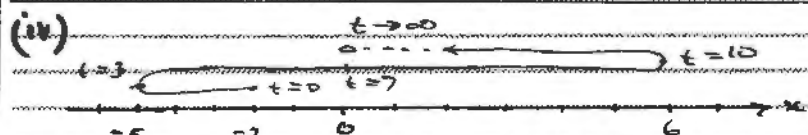
[1]

(ii) At greatest speed $\frac{dv}{dt} = 0$ and $\frac{dx}{dt} = \tan \theta$ $\therefore$ at approx 7 seconds ($t = 7$)

[1]

(iii) the particle approaches 0 from $x > 0$
at ever decreasing speed ($v < 0$ and $\ddot{x} > 0$)

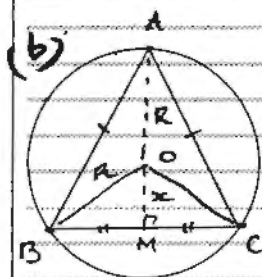
[1]



Total distance = $3 + \dots + 6$
 $= 20 \text{ m}$

$-2 \rightarrow -5 : 3 \text{ m}$
 $-5 \rightarrow 6 : 11 \text{ m}$
 $6 \rightarrow 0 : 6 \text{ m}$

[2]

(i) Area of $\triangle ABC$ $S = \frac{1}{2} BC \times h = \frac{1}{2} \times BC \times AM$.

$$R^2 = x^2 + BM^2 \quad (\text{Pyth. thm})$$

$$BM^2 = R^2 - x^2$$

$$BM = \sqrt{R^2 - x^2}$$

$$\therefore BC = 2BM = 2\sqrt{R^2 - x^2}$$

$$h = AM = x + R$$

$$\therefore S = \frac{1}{2} \times 2\sqrt{R^2 - x^2} \times (R + x)$$

$$\text{i.e. } S = (R + x)\sqrt{R^2 - x^2} \quad 0 \leq x \leq R$$

[2]

(ii) $\frac{dS}{dx} = 1 \cdot \sqrt{R^2 - x^2} + (R + x) \cdot \frac{1}{2} (R^2 - x^2)^{-\frac{1}{2}} \cdot (-2x)$

$$= \sqrt{R^2 - x^2} - \frac{x(R + x)}{\sqrt{R^2 - x^2}}$$

$$= \frac{R^2 - x^2 - Rx - x^2}{\sqrt{R^2 - x^2}}$$

$$\frac{dS}{dx} = \frac{R^2 - Rx - 2x^2}{\sqrt{R^2 - x^2}}$$

For possible max/min values of S to occur $\frac{dS}{dx} = 0$

$$\text{So } R^2 - Rx - 2x^2 = 0 \quad \text{only}$$

$$2x^2 + Rx - R^2 = 0$$

$$(2x - R)(x + R) = 0$$

$$x = -R \text{ or } \frac{1}{2}R$$

$$\text{So } x = \frac{1}{2}R \quad \checkmark \text{ as } x > 0 \text{ and } 0 \leq x \leq R$$

$$\therefore S = \frac{3R}{2} \sqrt{R^2 - R^2} = \frac{3\sqrt{2}}{4} R^2$$

continued $\rightarrow$

MATHEMATICS: Question 10 CONT.

Suggested Solutions

Marks

Marker's Comments

TEST nature at $x = \frac{1}{2}R$

METHOD 1

$R^2 - Rx - 2x^2 =$	x	0	$\frac{1}{2}R$	$\frac{1}{2}R$ or ...
$\frac{dS}{dx} = \frac{d}{dx}(R^2 - Rx - 2x^2)$	$\frac{dS}{dx}$	R	0	$-R^2/2$ (-)
$\frac{dS}{dx} = \frac{d}{dx}(R^2 - Rx - 2x^2)$		$(+)$		$\frac{dS}{dx} = \frac{d}{dx}(R^2 - Rx - 2x^2)$

$\therefore$ A Rel max TP at $x = \frac{1}{2}R$ ✓

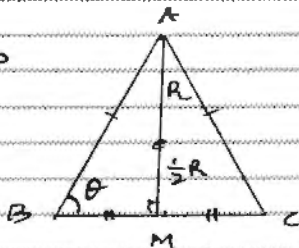
METHOD 2: $x=0$ $S=R^2$

$$x=\frac{R}{2} \quad S=\frac{3\sqrt{3}}{4}R^2 > R^2$$

$$x=R \quad S=0$$

and since S is continuous for $0 \leq x \leq R$
so Rel max TP is the absolute max at $x = \frac{1}{2}R$

Now



$$BM = \sqrt{R^2 - x^2} = \sqrt{R^2 - \frac{R^2}{4}} = \frac{R\sqrt{3}}{2}$$

$$\therefore BC = R\sqrt{3}$$

$$\tan \theta = \frac{R + \frac{1}{2}R}{\frac{R\sqrt{3}}{2}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

$$\therefore \theta = 60^\circ$$

$\therefore \angle C = \angle B = 60^\circ$ (equal angles opposite equal sides)
(data)

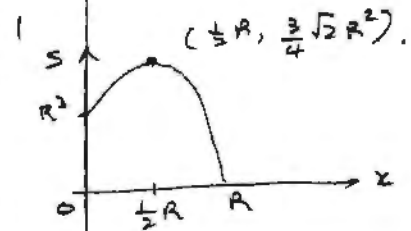
$$\therefore \angle A = 60^\circ$$

$\therefore$ max area when $x = \frac{1}{2}R$ and $\triangle ABC$ is equilateral.
qed

5

$$x=0.7R \quad \frac{dS}{dx} = -0.6 \times 1.7R^2 < 0$$

$$x=0.9R \quad \frac{dS}{dx} = -0.2 \times 1.9R^2 < 0$$



Ans 2: For
A proof that $\triangle ABC$
is equilateral
when $x = \frac{1}{2}R$